Whilst 71% of the area will be retained as either green or blue space, the expanded Campus will be delivered with commercial-led objectives. Wellcome intend to retain ownership of all assets on completion and this allows the design/procurement/delivery/handover/maintenance/management of all buildings to be driven by a holistic and standardised approach.

Urban&Civic was acquired by Wellcome Trust in 2021 and subsequently appointed as Development Manager to manage and deliver the development of the expansion under a Development Management Agreement. As part of this appointment Urban&Civic has sought to encourage and integrate a whole life approach to the development through engagement and consultation with stakeholders, designing in flexibility and innovation to the infrastructure and buildings to allow science and sustainability to flourish, whilst recognising the need to understand the estate and building management strategies following the completion of development.

Urban&Civic recognise the challenge of bringing forward a series of future-focussed sustainable buildings, incorporating both the latest science led technologies and innovative heating and cooling energy systems within environmentally sensitive and embodied carbon – lite structures. It therefore has sought to address this by taking the key stakeholders, namely Wellcome, their brokers and underwriters, on the journey through to completion of the development so all parties fully understand the risks and associated mitigation to be considered at each stage. The outputs would be led by the initial analysis of the key risks perceived by the stakeholder group.

The brief afforded by Wellcome as Landlord required the team to consider the following pre-requisites;

- Buildings were to be an attractive rental proposition from the outset.
- Long term flexibility of use was required to protect rental income and reputation.
- Property protection was to be considered over and above minimum requirements for life safety.
- There was a need to improve the useful life of a facility, optimisation of aftercare, life cycle costs and Post Occupancy Evaluation.
- Consideration of the need to improve the resale value of facilities.
- Recognition of Wellcome as an industry leader in the adoption of sustainable best practice.
- Buildings need to be insurable at a market-reasonable level.
- Wellcome recognised the need for close tenant engagement and establishing a framework of rules on risks and risk mitigation. Within this, Wellcome posed the following;
  1. Consider what limitations may be imposed on an occupational tenant use that may have high financial and reputational consequence.
  2. The building and the inbuilt systems being leased should protect the asset and its function.
  3. Recognise the potential for being able to obtain insurance for business Interruption risks

In October 2022, Urban&Civic appointed a team to respond to this brief. The assembled team each brought a key attribute and expertise to better respond to the brief. The team is as follows;

- Urban&Civic Projects Ltd – brief and project manager
- Arup – fire saftey consultant with unrivalled experience in designing successful, multi-disciplinary, integrated and sustainable building solutions.
- Waugh Thistleton – architect with a wealth of experience in detailing timber framed buildings.
- Glockling – previous director of the FPA and the RISC Authority with experience in researching, promoting, teaching, evaluating, certificating and developing tools, standards, product and methods in support of resilience within the insured commercial market – the project's 'critical friend'.
- AJGallagher – broker with a critical view on current market conditions and a high profile advocate of sustainable building design and insurance solutions.
- Lignum – experts in delivering insurance solutions from the Alternative Risk Transfer market.
- Imperial College – invited to comment on the general approach being taken by the study group.

The strategy and its outputs would consider the following three building types as the primary uses on the expanded development;

- Single Occupancy R&T, of the use type already present on the existing Campus,
- Multi-tenanted R&T/Incubator space, of the type already present on the existing Campus, and
- Low-rise residential blocks, not currently present on the existing Campus

Following initial dialogue with Wellcome and their broker, Marsh, the key and recognised risks of fire and water were confirmed as being relevant to the study. However, the matter of Estimated Maximum Loss (EML) was seen as the key driver due to the number of buildings being delivered across the expanded Campus.

The study therefore set about responding to these briefings by focussing on the delivery of two key documents;

- Design Guide – aA design guide prepared by Arup that sets out a novel calculation methodology to assess and limit the fire spread risk from one building to another, taking into consideration the anticipated fire service attendance times at the new Campus. It has been developed in recognition of existing methods in the UK not adequately capturing the hazard. The new methodology is validated against the latest research in masstimber fire dynamics and fire safety of timber cladding. The guide also includes recommendations to enhance resilience of suppression systems, and sets out tips to make building envelops and landscapes more resilient to wildfires. It is accompanied by illustrations by WTA of possible wall and roof construction methods that could be deployed on the site, and an example application of the external fire spread analysis..
- Design Guide Examples – a series of worked examples, using the RISC Authority Insurance Evaluation Tool, to allow design teams to apply the Design Guide to any of the three building types selected and based upon the illustrative master plan for the Wellcome Genome Campus expansion. This would also consider reducing EML below 100% within buildings and potential solutions for this.

The work has progressed over a twelve month cycle and has included extensive stakeholder engagement with Wellcome, their broker and the design teams appointed to bring forward the initial buildings at the expanded Wellcome Genome Campus. The two documents are seen to be consultative rather then instructional. They may naturally evolve during the course of the development, led by changing market perceptions, technological advances, new user classes to be considered or a significant change to the footprint of a specific building.

This volume constitutes the work undertaken in document 2 as noted above.